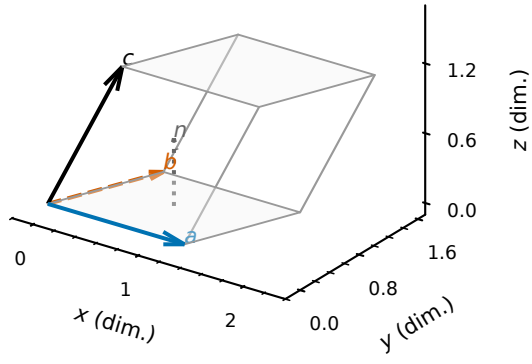
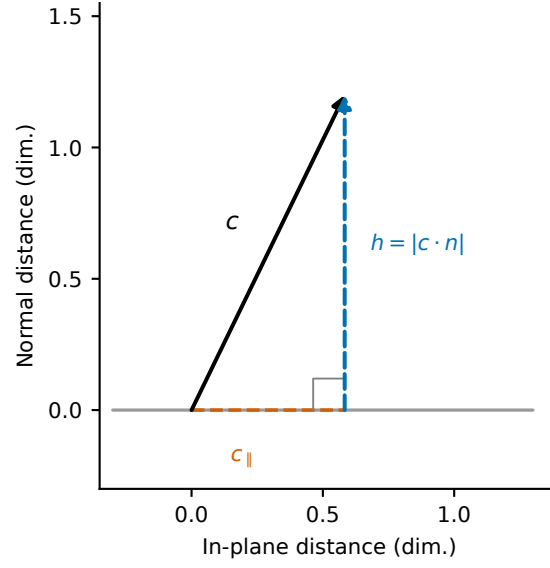
A



$A = \|a \times b\|;\ V = |(a \times b) \cdot c|$

B



Only normal height contributes: $V = A\,h$